Worksheet: Distributions & the Central Limit Theorem

DSCI401

Part 1: Geometric Distribution

Recall: if p is the probability of success on each independent trial,

$$P(\text{success on the } n\text{th trial}) = (1-p)^{n-1}p, \qquad \mu = \frac{1}{p}, \qquad \sigma = \sqrt{\frac{1-p}{p^2}}$$

1. A basketball player makes free throws 70% of the time. Assume each attempt is independent.

   (a) What is the probability that she makes her *first* free throw of the game on her 3rd attempt (i.e., misses the first two)?

   (b) What is the expected number of attempts until her first make? Interpret this value in a sentence.

   (c) Calculate the standard deviation of the number of attempts until her first make.

2. A quality control inspector finds that 4% of items coming off an assembly line are defective. Items are inspected one at a time until the first defective item is found.

   (a) What is the probability that the first defective item is the 5th one inspected?

   (b) How many items would the inspector expect to check before finding the first defective one?

Part 2: Binomial Distribution

Recall:

$$P(k \text{ successes in } n \text{ trials}) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \mu = np, \qquad \sigma = \sqrt{np(1-p)}$$

3. Continuing with the 70%-free-throw shooter: she attempts 8 free throws in a game.

   (a) What is the probability she makes *exactly* 6 of them? Show your work using the binomial formula (don't just state a decimal from software).

   (b) What is the expected number of free throws made, and the standard deviation?

   (c) Would it be unusual (more than 2 SD from the mean) for her to make only 3 of the 8? Justify your answer.

4. A 2023 survey found that 62% of adults in a certain country support a particular policy. Suppose you randomly sample 12 adults.

   (a) What conditions must be satisfied for this to be treated as a binomial setting? Check whether they're reasonably satisfied here.

   (b) Write, but do not necessarily evaluate, the expression for the probability that *at least* 10 of the 12 support the policy.

Part 3: Normal Approximation to the Binomial

Recall: the normal approximation to Binomial(n, p) is reasonable when $np \geq 10$ *and* $n(1-p) \geq 10$, using $\mu = np$ and $\sigma = \sqrt{np(1-p)}$.

5. Approximately 18% of adults in a large city are smokers. A researcher samples 400 adults at random.

   (a) Check whether the normal approximation conditions are satisfied.

   (b) Find μ and σ for the number of smokers in the sample.

   (c) Estimate the probability that *more than* 90 of the 400 sampled adults are smokers. Show your Z-score calculation.

   (d) Why might your answer to (c) be slightly off if you needed the probability of *exactly* 90 smokers instead of a range? What's the fix?

Part 4: Point Estimates, Sampling Distributions & the CLT

Recall: for a sample proportion $\hat{p}$ based on a sample of size n from a population with true proportion p, when observations are independent and $np \geq 10$, $n(1-p) \geq 10$:

$$\mu_{\hat{p}} = p, \qquad SE_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

6. A university claims that 72% of its graduates are employed within 6 months of graduation. Suppose this is the true population proportion, and you survey a random sample of 250 recent graduates.

   (a) Describe, in your own words, the difference between the *parameter* in this scenario and the *point estimate*.

   (b) Check the conditions for the Central Limit Theorem to apply.

   (c) Find $\mu_{\hat{p}}$ and $SE_{\hat{p}}$.

   (d) Estimate the probability that your sample's $\hat{p}$ comes out between 0.68 and 0.76. Show your Z-score work.

7. Suppose instead you only sample 15 graduates (still assuming true $p = 0.72$).

   (a) Check the success-failure condition for this sample size. What do you conclude?

   (b) Based on today's lecture, describe (without recalculating) how you'd expect the shape of the sampling distribution of $\hat{p}$ to look different for $n = 15$ compared to $n = 250$.

8. **Short answer:** In your own words, explain why a bigger sample size leads to a smaller standard error. Connect your answer to the formula $SE_{\hat{p}} = \sqrt{p(1-p)/n}$.